ALEXANDROS DELLIOS

alexandrosdellios@icloud.com | linkedin.com/in/alexandros-dellios | github.com/AlexandrosDellios

Summary

Master's student in Robotics at EPFL focused on robot learning and manipulation. My research goal is to help robots make sense of unstructured, real-world information so they can better assist people in everyday tasks.

Education

Master of Science (M.Sc.) in Robotics-Engineering *Expected–2028*
École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

Bachelor of Science (B.Sc.) in Micro-Engineering *2022-2025*
École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

Experience

Graduate Research Assistant | *BioRob* *July 2026 - Present*

- Researching egocentric data-driven robot manipulation, focusing on bridging the embodiment gap between human demonstrations and robotic execution.
- Developing foundation models to synthesize human egocentric data with physical robot teleoperation, improving policy robustness in unstructured environments.

Machine Learning Engineer | *EPFL AI Team* *Feb 2026 - Present*

- Contributed to the ground-up design and creation of a custom quadrupedal platform to act as a guide dog for the visually impaired, bridging the gap between hardware integration and intelligent control.
- Wrote the low-level CAN communication on the NVIDIA Jetson to ensure fast and reliable motor response.
- Addressing the challenge of varied-terrain navigation by building a simulation pipeline in NVIDIA Isaac Lab to train Reinforcement Learning policies.

Lead Design & Fabrication Engineer | *Mediacom EPFL* *Feb-Oct 2025*

- Won a competitive design pitch for the official EPFL Doctor Honoris Causa Award out of 12 proposals, taking full ownership of the project from concept to physical fabrication.
- Partnered with PhD researchers and the Mediacom team to execute the fabrication processes while ensuring the final product met strict design standards for the award's presentation to Marco Odermatt and others.

Projects

Deep-RL Quadruped Locomotion Controller | *Python/Pybullet/RL*

- Trained a deep learning model in a PyBullet simulation to teach a Unitree A1 quadruped to achieve stable locomotion on steep inclines (29% slopes) and high speeds.
- Tested multiple action spaces to find the most efficient setup, ultimately improving the robot's physical footstep accuracy by nearly 9%.

RL-Driven Glider Co-Design | *RL*

- Developed a Reinforcement Learning model to teach a glider how to execute highly precise landings, outperforming optimal control methods.
- Hardened the flight system against real-world conditions by training the model to seamlessly adapt to unpredictable wind disturbances and varying starting speeds.
- Engineered a simultaneous optimization process that allowed the agent to automatically redesign the physical wing shape while learning to fly, accelerating success convergence by 29.7%.

Image Processing for Uno Cards | *PyTorch / Torchvision*

- Engineered a lightweight (12M parameter) deep learning model using a customized ResNet-18 architecture and CBAM attention mechanisms to accurately detect and count overlapping Uno cards.
- Overcame real-world data scarcity by developing a synthetic data generation pipeline, applying aggressive visual augmentations (occlusion, perspective distortion) to prevent texture memorization and bridge the Sim2Real gap.
- Implemented a multi-stage fine-tuning strategy to successfully transition the model from synthetic data to real-world camera photos, ultimately achieving 75% accuracy on a Kaggle dataset.

MPC Rocket Landing Controller | *Python/CasADi*

- Designed a dual-layer flight control system (PID and MPC) for a simulated 6-DOF rocket to ensure highly accurate speed and trajectory tracking.
- Implemented a custom Nonlinear MPC algorithm to automatically correct complex aerodynamic drag and prevent orientation errors during landing.
- Engineered a safety-first landing protocol that guaranteed stability under unpredictable wind disturbances.

Technical Skills

Code : Python, C++, C, CUDA, Assembly
Frameworks & Libraries: ROS2 Jazzy, PyTorch, JAX, OpenCV, CasADi, Isaac Lab
Robotics & Control: Reinforcement Learning (PPO, SAC), Domain Randomization, Hardware-Software Co-Design, Model Predictive Control (MPC), Extended Kalman Filter, Deep Learning (Data Augmentation, Debiasing)
Simulators : Isaac Gym, Pybullet, MuJoCo
Hardware & embedded: NVIDIA Jetson (Orin Nano), STM32, ESP32, CAN bus, RTOS
CAD & Fabrication: Fusion 360, CATIA, KiCad, 3D Printing, Machining (Lathe/Milling)
Tools & DevOps: Linux, Git, TensorFlow, Docker, HPC / GPU Clusters, DeepSpeed